

4 The largest eigenvalue of an adjacency matrix of a random graph

Project proposed by: Ievgenii Afanasiev

Recommended software. Any software that has libraries for generating random values and computing eigenvalues of a symmetric matrix, e.g. Python.

Consider a random undirected graph with n vertices s.t. all the events

$$\{\text{Presence of the edge } (i, j)\}$$

are independent and have probability $\frac{p_n}{n}$. Such graph is called an Erdős–Rényi graph $G(n, \frac{p_n}{n})$. Its adjacency matrix is a symmetric matrix

$$A = (a_{ij}), \quad a_{ij} = \begin{cases} 1, & \text{if there is an edge connecting vertices } i \text{ and } j, \\ 0, & \text{otherwise.} \end{cases}$$

We would like to guess what is the asymptotic behavior, as $n \rightarrow \infty$, of the largest eigenvalue $\lambda_1(A)$ of A . Assuming $\mathbf{E}\lambda_1(A) \sim Cn^\alpha$, compute $\mathbf{E}\lambda_1(A)$ for several large values of n and determine C and α . Consider different dependencies of p_n on n :

1. $p_n = n$;
2. $p_n = an - b < n$;
3. $p_n = n^\beta, 0 < \beta < 1$;
4. $p_n = \log n$;
5. $p_n = \log \log n$;
6. $p_n = p = \text{const.}$

In Random Matrix Theory the entries of a random matrix are usually centered and normalized, s.t. the sum of variances in each row equals to 1:

$$M = \frac{1}{\sqrt{p_n}}(A - \mathbf{E}A).$$

The spectral behavior of such matrices is often universal, i.e. it does not depend on the distribution of the entries. Whether Bernoulli distribution is crucial in this case? Consider various random matrices M with other entry distributions having zero mean, unit variance and growing the fourth moment.

5 Integral values of rational sequences

Project proposed by: Przemysław Grabowski pgrabowski@kse.org.ua

Let $a_n, n \geq 1$ be a sequence of integers. Let $f_n : \mathbb{N}^n \rightarrow \mathbb{Q}$ be a sequence of functions.

$$\text{When } F_n = f_n(a_1, \dots, a_n) \in \mathbb{N}?$$

It is a super general form of a question I have asked in the Scottish cafe (Aug 2025), based on my first theorem from 2015.

Examples:

- $a_n = n, f_n(x_1, \dots, x_n) = x_n$: always
- $a_n = \min_{k \in \mathbb{N}} |k^2 - n|, f_n(x_1, \dots, x_n) = \frac{x_1 + \dots + x_n}{n}$: Grabowski 2015: $F_n = m, n, m \in \mathbb{N}$ if and only if $n = 9m^2 - 1, 9m^2 - 2m, 9m^2 + 2m$, see solution.
- It should be easy to start experimenting with this. Taking different nice sequences, other means. In particular, the previous item I did in high school, and it was experimental in nature.